

Muskan Batra

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SKILLS

- **Languages:** C, C++, Python
- **Technologies:** HTML, CSS, NumPy, Pandas
- **Databases/Tools:** MySQL
- **Soft Skills:** Problem solving, Team collaboration, Time management, Adaptability

PROJECTS

IoT-Based Smart Electricity Theft Detection

Dec 2025 - Jan 2026

- Engineered a real-time electricity theft detection system, integrating ESP32, ACS712 current sensor, and ZMPT101B voltage sensor to continuously monitor power consumption.
- Seamlessly integrated ThingSpeak for live data logging and a Telegram bot to instantly alert users of detected theft.
- Rigorously simulated and validated the complete circuit design in Proteus prior to implementation, minimizing hardware errors.
- Documented the project thoroughly with a technical report and presented it for End Term Evaluation (ETE).
- **Tech Stack:** ESP32, ACS712, ZMPT101B, ThingSpeak, Telegram Bot API, Proteus

Python for AI/ML — Self-Study Portfolio | GitHub

Jun 2026 - Present

- Structured and executed a 50-day self-learning roadmap, systematically progressing through Python foundations, NumPy, Pandas, visualization, and ML basics.
- Consistently maintained two GitHub repositories with daily commits — python-foundations for core Python and data-science-basics for Pandas/NumPy notebooks.
- Independently solved 20+ LeetCode problems, including Pandas-specific challenges such as Big Countries and Recyclable and Low Fat Products.
- **Tech Stack:** Python, NumPy, Pandas, Jupyter Notebook, Git/GitHub

TRAINING

Community Development Project (CDP) | Lovely Professional University

Jul 2025 - Aug 2025

Python Instructor — Local Community Outreach

- Designed and delivered a 6-session Python curriculum for local students in Sunam, covering print statements, variables, conditionals, loops, functions, and Turtle graphics.
- Independently prepared session materials, including a structured PowerPoint presentation, using a single laptop for hands-on demonstrations.
- Documented each session with geotagged photographic proof, ensuring compliance with program verification requirements.

CERTIFICATES

Python course for beginners | **Scalar**

Dec 2025

A1 level German language course | **LPU**

Aug 2026

The Primer - Design Thinking | **NPTEL**

Sep 2026

ACHIEVEMENTS

- **Completed** a full C programming course on Neocolab, strengthening core programming fundamentals.
- **Completed** a Database Management Systems (DBMS) course on Codetantra, covering SQL, normalization, and query optimization.
- **Achieved** an O grade in C, reflecting strong command over programming logic and syntax.
- **Achieved** an O grade in Mathematics in Semester 1 and Semester 2.
- **Independently solved** 20+ LeetCode problems as part of a self-driven 50-day Python/AI-ML learning plan.
- Maintained a CGPA of 9.22 in Year 1.

EDUCATION

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| Lovely Professional University | Phagwara, Punjab |
| • Bachelor of Technology - Computer Science and Engineering; CGPA: 9.22 | <i>Aug 2025 - Present</i> |
| Sant Attar Singh International School | Sangrur, Punjab |
| • Higher Secondary Education; Percentage: 80% | <i>May 2023 - Mar 2025</i> |
| Little Flower Convent School | Sangrur, Punjab |
| • Secondary Education; Percentage: 95.6% | <i>Jun 2023 - Mar 2025</i> |